

Future Atlas — Product Documentation

Prepared for: Clint
Product: Future Atlas (Global Pathways)
Version: Current build (Next.js 16 / React 19)
Platform: One Window
Date: August 2025

1. Executive Summary

Future Atlas is a **study-abroad planning platform** that helps students explore countries, universities, scholarships, costs, and eligibility through five AI-powered tools and a conversational AI companion. It is built as an embeddable, white-label product on the **One Window** platform — a study-abroad technology and services provider.

The interface is intentionally minimal, fast, and mobile-first. Students answer 2–3 guided questions per tool and receive personalized, AI-generated recommendations with match scores and rationale. The AI can also be opened as a full-screen chat for open-ended guidance.

Target users: Prospective international students (primarily Master's and Bachelor's applicants)
Primary value: Reduce decision paralysis; replace hours of scattered research with a guided, personalized workflow.

2. Platform & Technical Foundation

Layer	Technology
Core platform	One Window (hosted, managed, white-label)
Frontend framework	Next.js 16 (App Router), React 19, TypeScript
Styling	Tailwind CSS v4, shadcn/ui (base-nova), Lucide icons
Animation	Framer Motion
AI provider	Groq (Llama 3.3 70B) via serverless API route
AI features	Structured JSON output, client-side caching, conversation history
Deployment	Node.js server (next start), runs on port 3000/3001
Security headers	CSP <code>frame-ancestors</code> (configurable via <code>EMBED_ALLOWED_ORIGINS</code>), <code>X-Frame-Options: SAMEORIGIN</code> on protected routes

What "built on One Window" means:
The core infrastructure (auth, data, AI orchestration, embedding, partner integrations) is provided by One Window. Future Atlas is a configured product instance — not a custom build from scratch. Customization is at the UI/UX, prompt, and workflow level.

3. User Interface Overview

3.1 Global Layout

- Background:** Light neutral (#F8F9FC) across all pages
- Header (sticky, blurred):** Back button + logo ("Future Atlas / Global Pathways") — clicking home clears temporary session state
- Max content width:** max-w-7xl (1280px) with responsive padding
- Typography:** Geist Sans / Geist Mono (Google Fonts)
- Footer note:** "Your journey. Your choices. Your future." + "Powered by One Window" on tools

3.2 Home / Dashboard (/)

The landing page is the **tool hub** — five cards in a responsive grid (1 featured + 4 standard):

Tool	Tag	Route	Purpose
Explore Countries	Discover	/countries	Compare destinations by study field, budget, priorities
Explore Universities	Find your fit	/universities	Match universities to course, budget, profile
Scholarship Explorer	Save money	/scholarships	Discover funding opportunities
Cost Calculator	Plan your budget	/cost-calculator	Estimate full study-abroad budget
Eligibility Checker	Check your profile	/eligibility	Quick academic/English readiness assessment

AI Section (prominent): Dark card with "Ask Future Atlas AI" CTA → opens full-screen mentor chat modal.

Guidance CTA: "Ready for personalized guidance?" → links to /guidance (partner lead capture).

3.3 Tool Pages (Shared Pattern)

Each tool page (/countries, /universities, /scholarships, /cost-calculator) follows:

Header
└─ Tool Component (interactive, multi-step or single-view)
└─ Guidance CTA (fixed at bottom)

Country Explorer (/countries) — 3-step guided flow

- What to study** — 6 preset fields + "Other" (custom input)
- Budget range** — 4 bands + "No Preference" + "Other"
- Top priority** — Career/PR/Low Cost/Rankings/Scholarships + "Other"

→ **Results:** 3 country cards with:

- Match score (70–98%)
- 2-letter code badge

- Description tailored to answers
- Tags (e.g., "Post-study work", "Low tuition")
- "Best overall match" label on #1

University Explorer (/universities) — Same 3-step pattern

Fields: Study area, Budget, Priority
Results: 3 university cards with match score, location, tuition band, type, highlights

Scholarship Explorer (/scholarships) — Same 3-step pattern

Fields: Study level, Destination, Profile strength
Results: 3 scholarship matches with provider, country, amount, coverage, deadline guidance, match score

Cost Calculator (/cost-calculator) — Single view with AI

- Input: Country, Study level, Duration, Lifestyle preference
- Output: Structured breakdown (tuition, accommodation, living, insurance, visa, travel, other, total) + budget status label + AI analysis text
- All estimates shown as ranges with assumptions noted

Eligibility Checker (/eligibility) — Direct AI chat

Opens full-screen Future Atlas AI in `eligibility` mode immediately (no multi-step form).
Returns: Score (0–98), status label, summary, 3-category breakdown (Academic / English / Programme Fit), 3 next steps.

3.4 Future Atlas AI (Conversational Companion)

Access points:

- Home dashboard → "Ask Future Atlas AI" (mentor mode)
- Embed view → same
- Eligibility page → auto-opens in eligibility mode
- Each tool has a mode switcher in the AI header dropdown

Modes (6):

Mode	Label	Use Case
mentor	Study Abroad Mentor	General Q&A, open-ended guidance
country	Country Explorer	Destination recommendations
university	University Explorer	University matching
scholarship	Scholarship Explorer	Funding discovery
cost	Cost Calculator	Budget estimation
eligibility	Eligibility Checker	Profile assessment

UI Features:

- Full-screen modal (desktop) / sheet (mobile)
- Mode selector dropdown (shows 4 primary, "More" expands)
- Streaming "Thinking..." indicator
- Markdown rendering (lists, bold, headings)
- Suggestion chips on first message
- Client-side response caching (24hr TTL, 250-entry LRU)
- Disclaimer: "Future Atlas AI can make mistakes. Always verify... Powered by One Window"

Technical: Calls `/api/ai` (POST) with mode, message, history, optional structured inputs. Groq Llama 3.3 70B with JSON schema enforcement for structured modes.

3.5 Embed Mode (/embed)

Purpose: White-label iframe embed for partner sites (university portals, agent platforms, etc.)

Flow:

1. Parent site embeds `/embed` in iframe
2. Server validates `Referer/Origin` against `EMBED_ALLOWED_ORIGINS` env var
3. If allowed → renders tool selector grid (same 5 tools + AI)
4. Tool selection → `/embed?tool=countries` (etc.) renders that tool full-width
5. "Back to tools" button returns to grid
6. AI opens in mentor mode within embed

Security: CSP `frame-ancestors` restricted to allowed origins. Unauthorized attempts logged.

3.6 Guidance Page (/guidance)

Lead capture / partner handoff page. (Implementation not reviewed — route exists, linked from CTA.)

4. Key User Flows

Flow A: First-time visitor (Country Explorer)

1. Lands on / → sees 5 tool cards + AI prompt
2. Clicks "Explore Countries" → `/countries`
3. Step 1: Selects "Computer Science" → Continue
4. Step 2: Selects "\$15,000–\$30,000" → Continue
5. Step 3: Selects "Career & Jobs" → "Find My Countries"

- 6. Sees 3 country cards with match scores, descriptions, tags
- 7. Clicks "Change preferences" or opens AI for follow-up

Flow B: Returning user (Embed)

- 1. Partner site loads /embed in iframe
- 2. User sees tool grid, clicks "Cost Calculator"
- 3. Enters country + preferences → sees structured cost breakdown
- 4. Opens AI → asks "How much for Germany vs France?" → gets comparison
- 5. Closes embed → returns to partner site

Flow C: Quick eligibility check

- 1. Navigates to /eligibility
- 2. AI chat opens immediately in eligibility mode
- 3. Types: "I have 72% in CS, want Master's in Germany, IELTS 7.0"
- 4. Receives: Score (e.g., 84), status "Strong initial profile", breakdown, 3 next steps

5. AI Behavior & Guardrails

Aspect	Detail
Model	Llama 3.3 70B (Groq)
Temperature	0.7
Structured modes	Enforced JSON schema via response_format: json_object
System prompts	Defined in /lib/ai/personalities.ts — role, tone, scope
Routing instructions	Injected into every request — keeps AI focused on 5 tools
Scope filter	Mentor mode rejects non-study-abroad topics (celebrity, crypto, politics, etc.)
Caching	Client-side (localStorage, 24hr) + server in-memory (24hr, 250 entries)
Deduplication	Pending request map prevents duplicate Groq calls
Rate limit handling	Returns 429 with friendly message
Error handling	Graceful fallback: "I couldn't connect to the AI right now"
Disclaimer	Shown in every AI response footer

What the AI does NOT do:

- Guarantee admission, visa approval, or scholarship awards
- Access real-time university databases or live tuition data
- Store personal data beyond session cache
- Answer non-study-abroad questions (in mentor mode)

6. Data & Privacy

Data Type	Storage	Retention
Tool selections (country, university, etc.)	localStorage (keys: future-atlas-*)	Cleared on "Clear Selection" or home click
AI conversation history	In-memory (component state) + client cache	24hr TTL, cleared on modal close
AI response cache	Client localStorage + server Map	24hr / 250-entry LRU
Embed access logs	Server logs (unauthorized attempts)	Standard log retention

No backend database in this build — all ephemeral. Production One Window deployments add persistent user profiles, saved searches, and partner CRM sync.

7. Configuration (Environment Variables)

Variable	Purpose	Required
GROQ_API_KEY	AI provider authentication	Yes
EMBED_ALLOWED_ORIGINS	Comma-separated list of origins allowed to iframe /embed	For embed
NEXT_PUBLIC_*	None currently used	No

8. What's Not In Scope (This Build)

- User authentication / accounts
- Saved searches / history across sessions
- Real university data APIs (AI generates from training knowledge)
- Live tuition/visa feeds
- Partner CRM integration
- Multi-language (UI is English only)
- Admin dashboard / analytics
- Payment / subscription logic

These are platform-level capabilities available via One Window in production deployments.

9. Roadmap Considerations (Discussed)

Area	Status
Real data integration	Planned — connect university/scholarship APIs to ground AI outputs
User accounts + saved journeys	Platform feature — enable via One Window config
Agent/university white-label	Embed mode is the foundation — theming/config next

Multi-step eligibility form	Current chat-only; guided form option under review
Cost calculator granularity	Add city-level, accommodation type, dependents
Analytics / funnel tracking	Planned via One Window events

10. Quick Reference: Routes

Route	Purpose	Auth
/	Dashboard (tool hub)	Public
/countries	Country Explorer	Public
/universities	University Explorer	Public
/scholarships	Scholarship Explorer	Public
/cost-calculator	Cost Calculator	Public
/eligibility	Eligibility Checker (AI chat)	Public
/embed	Embed tool selector	Origin-validated
/embed?tool=*	Embedded single tool	Origin-validated
/guidance	Partner lead capture	Public
/api/ai	AI endpoint (POST)	Public (rate-limited)

11. Support & Escalation

- **Platform issues (uptime, embed, One Window config):** One Window support
- **AI quality / prompts:** Product team (adjust personalities in `/lib/ai/personalities.ts`)
- **UI/UX changes:** Frontend team (Next.js / Tailwind / shadcn)
- **Security headers / CSP:** DevOps / platform config

End of documentation. This covers the product as currently built. For implementation details (code paths, component props, API schemas), see the source repository.